

HYUNGJOON KOO

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Associate Professor, Sungkyunkwan University, Department of Computer Science and Engineering,
College of Computing and Informatics Sep 2024 – Present

Assistant Professor, Sungkyunkwan University, Department of Computer Science and Engineering,
College of Computing and Informatics Feb 2021 – Aug 2024

EDUCATION

Postdoc., Georgia Tech, School of Computer Science, College of Computing Jun 2019 – Dec 2020

- Adviser: Taesoo Kim (Systems Software & Security Lab)
- Research Area: Software Security, Artificial Intelligence for Security

Ph.D., Stony Brook University, Department of Computer Science Aug 2013 – May 2019

- Adviser: Michalis Polychronakis (Hexlab)
- Research Area: Binary Protection, Software Diversification against Code Reuse Attacks
- Thesis: Practical Software Specialization against Code Reuse Attacks

M.Sc., Korea University, Information Management and Security Mar 2008 – Feb 2010

- Adviser: Sangjin Lee (Digital Forensic Lab)
- Thesis: Pre-detection Model for Trusted Insider's Leaks and Manipulation from a Forensic Perspective

B.Sc., Hanyang University, Industrial Engineering Mar 1998 – Aug 2005

- Graduated with College Honors Cum Laude

CONFERENCES

Note that [*] represents either the first author or the (co-) corresponding author.

- Fool Me If You Can: On the Robustness of Binary Code Similarity Detection Models against Semantics-preserving Transformations (To appear), Jiyong Uhm, Minseok Kim, Michalis Polychronakis, and **Hyungjoon Koo**, *In Proceedings of the The ACM International Conference on the Foundations of Software Engineering (FSE '26)*, 2026 [*]
- UnPII: Unlearning Personally Identifiable Information with Quantifiable Exposure Risk (To appear), Intae Jeon, Yujeong Kwon, and **Hyungjoon Koo** *In Proceedings of the Software Engineering in Practice track of the International Conference on Software Engineering (ICSE-SEIP '26)*, 2026 [*]
- A Deep Dive into Function Inlining and its Security Implications for ML-based Binary Analysis (To appear), Omar Abusabha, Jiyong Uhm, Tamer Abuhmed, and **Hyungjoon Koo**, *In Proceedings of the Network and Distributed System Security Symposium (NDSS '26)*, 2026 [*]
- Rescuing the Unpoisoned: Efficient Defense against Knowledge Corruption Attacks on RAG Systems, Minseok Kim, Hankook Lee, and **Hyungjoon Koo**, *In Proceedings of the Annual Computer Security Applications Conference (ACSAC '25)*, 2025 [*]
- A Decade-long Landscape of Advanced Persistent Threats: Longitudinal Analysis and Global Trends, Shakhzod Yuldoshkhuaev, Mijin Jeon, Doowon Kim, Nick Nikiforakis, and **Hyungjoon Koo**, *In the 32nd ACM Conference on Computer and Communications Security (CCS'25)*, 2025 (*Distinguished Paper Award*) [*]
- BOOTKITTY: A Stealthy Bootkit-Rootkit Against Modern Operating Systems (To appear) Junho Lee, Jihoon Kwon, HyunA Seo, Myeongyeol Lee, Hyungyu Seo, Jinho Jung, and **Hyungjoon Koo**, *In the 19th USENIX WOOT Conference on Offensive Technologies (WOOT'25)*, 2025
- Evaluating the Effectiveness and Robustness of Visual Similarity-based Phishing Detection Models (To appear), Fujiao Ji, Kiho Lee, **Hyungjoon Koo**, Wenhao You, Euijin Choo, Hyoungshick Kim, and Doowon Kim, *In the 34th USENIX Conference on Security Symposium (USENIX'25)*, 2025
- An Empirical Study of Black-box based Membership Inference Attacks on a Real-World Dataset, Yujeong Kwon, Simon S. Woo, and **Hyungjoon Koo**, *In the 17th International Symposium on Foundations and Practice of Security (FPS 2024)*, 2024 [*]
- R2I: A Relative Readability Metric for Decompiled Code, Haeun Eom, Dohee Kim, Sori Lim, **Hyungjoon Koo**, and Sungjae Hwang, *In the ACM International Conference on the Foundations of Software Engineering (FSE '24)*, 2024 [*]
- BinAdapter: Leveraging Continual Learning for Inferring Function Symbol Names in a Binary (To appear), Nozima Murodova and **Hyungjoon Koo**, *In the 19th ACM ASIA Conference on Computer and Communications Security (ASIACCS '24)*, 2024 [*]

- **BENZENE: A Practical Root Cause Analysis System with an Under-Constrained State Mutation**, Younggi Park, Hwiwon Lee, Jinho Jung, **Hyungjoon Koo**, and Huy Kang Kim, *In the 45th IEEE Symposium on Security & Privacy (S&P '24)*, 2024 (*Distinguished Paper Award*) [*]
- **A Transformer-based Function Symbol Name Inference Model from an Assembly Language for Binary Reversing**, Hyunjin Kim, Jinyeong Bak, Kyunghyun Cho, and **Hyungjoon Koo**, *In the 18th ACM Asia Conference on Computer and Communications Security (ASIACCS '23)*, 2023 [*]
- **SmartMark: Software Watermarking Scheme for Smart Contracts**, Taeyoung Kim, Yunhee Jang, Chanjong Lee, **Hyungjoon Koo**, and Hyounghshick Kim, *In the 45th IEEE/ACM International Conference on Software Engineering (ICSE '23)*, 2023 [*]
- **Practical Binary Code Similarity Detection with BERT-based Transferable Similarity Learning**, Sunwoo Ahn, Seonggwon Ahn, **Hyungjoon Koo**, and Yunheung Paek, *In the 38th Annual Computer Security Applications Conference (ACSAC '22)*, 2022 [*]
- **DeView: Confining Progressive Web Applications by Debloating Web APIs**, ChangSeok Oh, Sangho Lee, Chenxiong Qian, **Hyungjoon Koo**, and Wenke Lee, *In the 38th Annual Computer Security Applications Conference (ACSAC '22)*, 2022 [*]
- **IoTivity Packet Parser for Encrypted Messages in Internet of Things** Hyeonah Jung, **Hyungjoon Koo**, and Jaehoon (Paul) Jeong. *In the 24th International Conference on Advanced Communications Technology (ICACT '22)*, 2022 [*]
- **A Look Back on a Function Identification Problem** **Hyungjoon Koo**, Soyeon Park, and Taesoo Kim. *In the 37th Annual Computer Security Applications Conference (ACSAC '21)*, 2021 [*]
- **Software Watermarking via a Binary Function Relocation** Honggoo Kang, Yonghwi Kwon, Sangjin Lee, and **Hyungjoon Koo**. *In the 37th Annual Computer Security Applications Conference (ACSAC '21)*, 2021 [*]
- **Slimium: Debloating the Chromium Browser with Feature Subsetting**, Chenxiong Qian, **Hyungjoon Koo**, Changseok Oh, Taesoo Kim, and Wenke Lee. *In the 27th ACM Conference on Computer and Communications Security (CCS '20)*, 2020
- **Compiler-assisted Code Randomization**, **Hyungjoon Koo**, Yaohui Chen, Long Lu, Vasileios P. Kemerlis, and Michalis Polychronakis. *In the 39th IEEE Symposium on Security & Privacy (S&P '18)*, 2018 *Top 10 Finalist, Cyber Security Awareness Week (CSAW '18)*, 2018 [*]
- **Defeating Zombie Gadgets by Re-randomizing Code Upon Disclosure**, Micah Morton, **Hyungjoon Koo**, Forrest Li, Kevin Z. Snow, Michalis Polychronakis, and Fabian Monrose. *In the 9th International Symposium on Engineering Secure Software and Systems (ESSoS '17)*, 2017
- **Return to the Zombie Gadgets: Undermining Destructive Code Reads via Code-Inference Attacks**, Kevin Z. Snow, Roman Rogowski, Jan Werner, **Hyungjoon Koo**, Fabian Monrose, and Michalis Polychronakis. *In the 37th IEEE Symposium on Security & Privacy (S&P '16)*, 2016
- **Juggling the Gadgets: Binary-level Code Randomization using Instruction Displacement**, **Hyungjoon Koo** and Michalis Polychronakis. *In the 11th ACM Asia Conference on Computer and Communications Security (ASIACCS '16)*, 2016 [*]
- **Identifying Traffic Differentiation in Mobile Networks**, Arash Molavi Kakhki, Abbas Razaghpanah, Anke Li, **Hyungjoon Koo**, Rajeshkumar Golani, David Choffnes, Phillipa Gill, and Alan Mislove. *In the 15th ACM Internet Measurement Conference (IMC '15)*, 2015

JOURNALS

Note that [*] represents either the first author or the (co-)corresponding author.

- **ToolPhet: Inference of Compiler Provenance from Stripped Binaries with Emerging Compilation Toolchains**, Nozima Murodova Hohyeon Jang, and **Hyungjoon Koo**, *IEEE Access*, vol. 11, pp. 12667 - 12682, doi: 10.1109/ACCESS.2024.3355098, 2024 [*]
- **Demystifying the Regional Phishing Landscape in South Korea**, Hyunjun Park, Kyungchan Lim, Doowon Kim, Donghyun Yu, and **Hyungjoon Koo**, *IEEE Access*, vol. 11, pp. 130131 - 130143, doi: 10.1109/ACCESS.2023.3333883, 2023 [*]
- **Binary Code Representation with Well-balanced Instruction Normalization**, **Hyungjoon Koo**, Soyeon Park, Daejin Choi, and Taesoo Kim, *IEEE Access*, vol. 11, pp. 29183 - 29198, doi: 10.1109/ACCESS.2023.3259481, 2023 [*]

WORKSHOPS, POSTERS

- **On the Learnability, Robustness, and Adaptability of Deep Learning Models for Obfuscation-applied Code**, Jiyong Uhm, Yujeong Kwon, and **Hyungjoon Koo**, *Workshop on Software Understanding and Reverse Engineering (SURE '25; Co-located with CCS '25)*, 2025

- Evaluating Password Composition Policy and Password Meters of Popular Websites, Kyungchan Lim, Joshua Hankyul Kang, Matthew Dixon, **Hyungjoon Koo**, and Doowon Kim, *Workshop on Designing Security for the Web (SecWeb '23; Co-located with S&P '23)*, 2023
- Inference of Compiler Provenance from Malware, Hohyun Jang and **Hyungjoon Koo**, *Poster in the the 22nd World Conference on Information Security Applications (WISA '21)*, 2021 [*]
- Semantic-aware Binary Code Representation with BERT, **Hyungjoon Koo**, Soyeon Park, Daejin Choi and Taesoo Kim (ArXiv), 2021 [*]
- Configuration-Driven Software Debloating, **Hyungjoon Koo**, Seyedhamed Ghavamnia, and Michalis Polychronakis. *In the 12th European Workshop on Systems Security (EuroSec '19; Co-located with EuroSys '19)*, 2019 [*]
- The Politics of Routing: Investigating the Relationship between AS Connectivity and Internet Freedom, Rachee Singh, **Hyungjoon Koo**, Najmehalsadat Miramirkhani, Fahimeh Mirhaj, Leman Akoglu, and Phillipa Gill. *In the 6th USENIX Workshop on Free and Open Communications on the Internet (FOCI '16)*, 2016

WORK EXPERIENCES

- Research Assistant, Stony Brook University** May 2014 – May 2019
- System / Software Security (Michalis Polychronakis)
 - Traffic Differentiation / Internet Censorship (Philipa Gill)
- Intern, Fujitsu Laboratories of America** Jun 2018 – Aug 2018
- Teaching Assistant, Stony Brook University** Aug 2013 – Dec 2017
- [CSE102] Introduction to Web Design and Programming (Ahmad Esmaili), Fall 2013
 - [CSE130] Introduction to Programming in C (Ahmad Esmaili), Fall 2013
 - [CSE312] Legal, Social, and Ethical Issues in Information Systems (Robert Johnson), Spring 2014
 - [CSE408] Network Security (Undergraduate level; Robert Johnson), Spring 2014
 - [CSE508] Network Security (Graduate level; Michalis Polychronakis), Fall 2017
- Intern, Fujitsu Laboratories of America** Jun 2016 – Aug 2016
- Lecturer** Mar 2013 – Jul 2013
- Security Essentials, Korea Productivity Center, July 2013
 - Network Security for Rwanda government officials, KISA, Mar 2013
- Security Researcher at Security Compliance Team, Shinhan Bank** Jul 2011 – Sep 2012
- Assistant Manager at Information Security Team, Samsung SDS** Jan 2006 – Jul 2011

PROFESSIONAL ACTIVITIES

- Committee Services**
- General Chair in Information Security Conference (ISC 2025)
 - Program Committee in the Workshop on Large AI Systems and Models with Privacy and Security Analysis (Co-located with CCS 2025)
 - Program Committee in the 1st Workshop on Software Understanding and Reverse Engineering (Co-located with CCS 2025)
 - Program Committee in 8th Deep Learning Security and Privacy Workshop (Co-located with S&P 2025)
 - Program Co-chair for the Security & Privacy Domain in International Conference on Parallel and Distributed Systems (ICPADS 2024)
 - Program Committee in IEEE Symposium on Security and Privacy (S&P 2024)
 - Organizing Committee in IEEE Secure Development Conference (SecDev 2023-24)
 - Program Committee in Annual Computer Security Applications Conference (ACSAC 2023-25)
 - Program Committee in International Symposium on Foundations & Practice of Security (FPS 2022-25)
 - Program Committee in NYU's Cybersecurity Games and Conference (CSAW 2019-2024)
- Journal Services as a Reviewer or an Editor**
- The Journal of Supercomputing (2025)
 - Korea Institute of Information Security & Cryptology (KIISC, 2024-25)
 - ACM Transactions on Software Engineering and Methodology (TOSEM, 2024)
 - IEEE Transactions on Software Engineering (TSE, 2024)
 - Editor, Journal of Information Processing Systems (JIPS, 2024-25)
 - Neurocomputing (2024)

- Information and Software Technology (IST, 2024)
- IEEE Transactions on Dependable and Secure Computing (TDSC, 2022)
- Computers and Security (COSE, 2023, 2021)
- IEEE Internet Computing (IC, 2022)
- International Journal of Information Security (IJIS, 2020-21)
- IEEE Security & Privacy Magazine (S&P, 2019-21, 2024)
- Frontiers of Information Technology & Electronic Engineering (FITEE, 2020, 2024)
- International Journal of Information Security (IJIS, 2020)
- IEEE Access (2019)
- IEEE/ACM Transactions on Networking (TON, 2019)

Patents

- Device and Method for Program Behavior Generation Using Under-Constrained State Mutation for Automated Root Cause Analysis, Younggi Park, Hwiwon Lee, Huykang Kim, **Hyungjoon Koo**; (KR) App No. 10-2025-0065904
- Relative readability index for Decompiled Codes, **Hyungjoon Koo**, Sungjae Hwang, Haeun Eom, Dohee Kim, Sori Lim; (KR) App No. 10-2024-0105125
- Methods and apparatus for inferring function symbols from assembly code in transformer-based executable binary, and recording medium, **Hyungjoon Koo**, Hyunjin Kim, and Jinyeong Bak, (KR) Patent No. 10-2781464; (US) App No. 18-674-646
- Watermarking method for smart contract, Hyoungshick Kim, Taeyoung Kim, Yunhee Jang, Chanjong Lee, and **Hyungjoon Koo**, (KR) Patent No. 10-2702323, (US) App No. 18-368-734
- Apparatus and method for detecting similarity of binary code, Sunwoo Ahn, Seonggwon Ahn, **Hyungjoon Koo**, and Yunheung Paek, (KR) App No. 10-2023-0035069; (US) App No. 18-596-194
- Method and device of embedding watermark in software, **Hyungjoon Koo**, Sangjin Lee, and Honggu Kang, (KR) App No. 10-2023-0035069
- Method of detecting intrusion for infotainment systems and apparatus thereof, **Hyungjoon Koo**, (KR) Patent No. 10-2022-0005421
- System and method for responding DDoS offensive, Changryul Huh, Bonjae Koo, Bonghui Park, **Hyungjoon Koo**, Kyutae Jeong, (KR) Patent No. 10-2010-0065260

Awards/Grants

- Distinguished Paper Award (Title; BENZENE: A Practical Root Cause Analysis System with an Under-Constrained State Mutation) at IEEE Symposium on Security and Privacy 2024 (May 2024)
- Best Paper Award (Title; RoBERTa-based Obfuscated Binary Code Similarity Detection) at Conference on Information Security and Cryptography-Summer 2024 (Jun. 2024)
- Best Paper Award (Title; A Study of Executable Binaries with Emerging Compilation Tools) at Conference on Information Security and Cryptography-Winter 2024 (Jun. 2022)
- Student Grant for the 26th USENIX Security Symposium in Vancouver (Aug. 2017)

Invited Talks

- Executable Binary Analysis with AI, Workshop on Information Security and Cryptography (WISC '23) (Sep. 2023)
- A Transformer based Function Symbol Name Inference Model from an Assembly Language for Binary Reversing, Kyunghee University Seminar (Aug. 2023)
- Understanding of Advanced Code Reuse Attacks and Defenses, UNIST Seminar (Dec. 2022)
- A Practical Binary Similarity Detection Approach, Sejong Cybersecurity Workshop 0x01 (Oct. 2022)
- Binary Code Representation for Deep Learning and its Applications, Software Convergence Symposium (SWCS '22) (Apr. 2022)
- Semantic-aware Binary Code Representation with Deep Learning, Fall KAIST Security Colloquium (Nov. 2021)
- Executable Binary Code Representation with Deep Learning, In the 21ST KOCSEA Technical Symposium Program (Nov. 2021)
- Crash Course on Deep Learning for Security, Soongsil University (Jul. 2021)
- Toward (Better) Binary Code Representation with Deep Learning, Seoul National University (Jun. 2021)
- Software Protection via Code Randomization, University of Tennessee (Nov. 2020)
- Practical Software Specialization against Code Reuse Attacks, Sungkyunkwan University and KAIST (Feb. 2019)

- Practical Software Hardening against Code Reuse Attacks, Georgia Tech (Nov. 2018)
- Software Hardening with Code Diversification, CS Colloquium at SUNY Korea (Jun. 2018), Korea University and Samsung Research (May 2018)
- Software Hardening, Cyber Symposium by the Stony Brook Computing Society (Apr. 2018)
- Elaborate Attacks with Existing Tools, National Computing & Information Agency (May 2013)
- Anonymizing Yourself with Tor, Korea Internet & Security Agency (Apr. 2013)

Invited Lectures

- Binary Reversing with AI, Lecture Series on AI Applications by KIISC, Korea Institute of Information Security and Cryptography (Aug. 2024)
- Software Security, Seoul Science High School (Jan. 2022-2024)
- Security with AI, Kepco KDN (Nov. 2021)

Translation of Technical Books/Articles into Korean

- Gray Hat C# (ISBN: 1593277598, 2018)
- Logging and Log Management (ISBN: 1597496359, 2014)
- Practical Malware Analysis (ISBN: 1593272901, 2013)
- Malware Analyst's Cookbook and DVD (ISBN: 0470613033, 2011)
- Cryptography Engineering (ISBN: 0470474246, 2010)
- OWASP Top10, SANS Top20, and ISM Top10 (2007, 2010)

Write-ups

- Keychain Analysis for Mac OS X, Kyeongsik Lee and **Hyungjoon Koo** (2013)
- Hunting OS X Rootkit in Memory, Kyeongsik Lee, Jinkook Kim, and **Hyungjoon Koo** (2013)
- A Guidebook for Building and Operating CERT by KISA (2007)